

EXAM

Video Processing
Course number 0512-4263-01
Spring 2020
Faculty of Engineering
Tel-Aviv University
Prof. Shai Avidan

DATE: 8/7/2020
TIME: 9am
DURATION: 3 hours

Read all of the following information before starting the exam:

- Show all work, clearly and in order.
- Justify your answers.
- Please keep your written answers brief; be clear and to the point.
- Each answer should take one page only (text, images and equations).
- All text should be typed with font size 11.
- This exam has 8 questions and is worth 100 points.
- Good luck!

1. (13 points) OPTIC FLOW

Cycle-consistency states that the flow from pixel p to pixel q should equal the (negative) flow from q to p , where p is a pixel in the first image and q is a pixel in the second image. Modify the Horn-Schunck equations to encode the cycle-consistency prior.

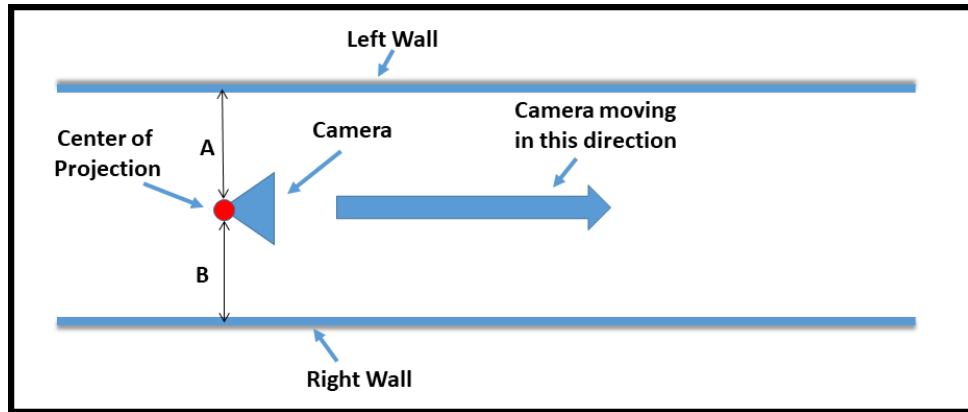


Figure 1: Top view diagram. A camera (the triangle) is moving in the direction of the arrow straight down the corridor. Let A denote the distance of the camera from the left wall, and let B denote the distance from the right wall.

2. (12 points) OPTIC FLOW

A robot equipped with a forward facing camera is moving down a straight and long corridor (see Figure 1). You also know that the floor, walls and ceiling of the corridor are textured, so optic flow algorithms work well. Let A denote the distance from the camera center of projection to the left wall and let B denote the distance from the camera center of projection to the right wall. Suggest a way to use optic flow to determine if $A = B$?



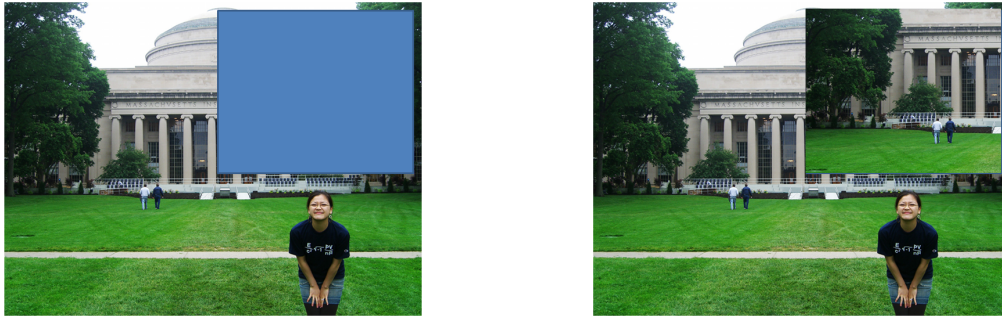


Figure 2: (Left) Image with hole to be filled. (Right) Result of hole filling.

3. (13 points) IMAGE INPAINTING

Consider Figure 2. The user tried to inpaint the marked rectangle using the Bidirectional similarity measure and got this result. Explain how this can happen and propose a solution to fix this problem.



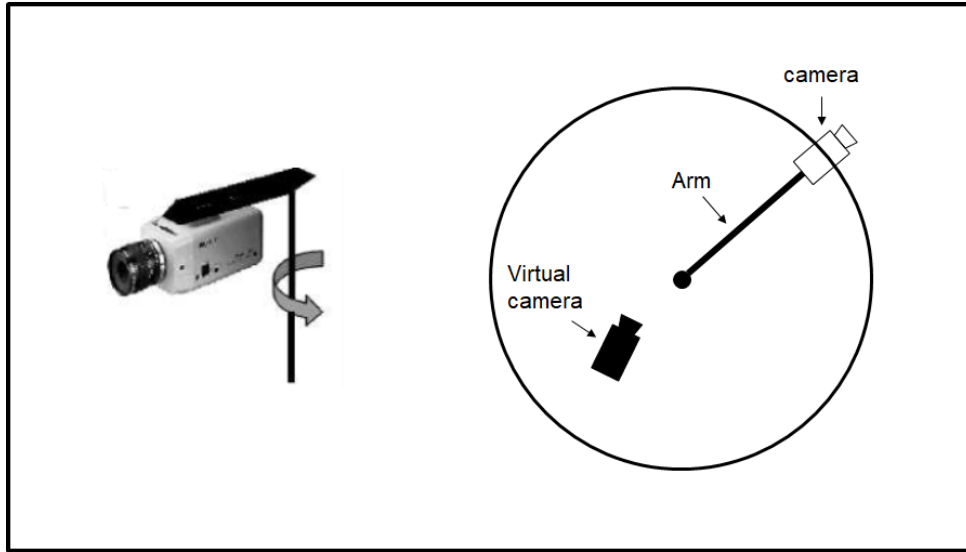
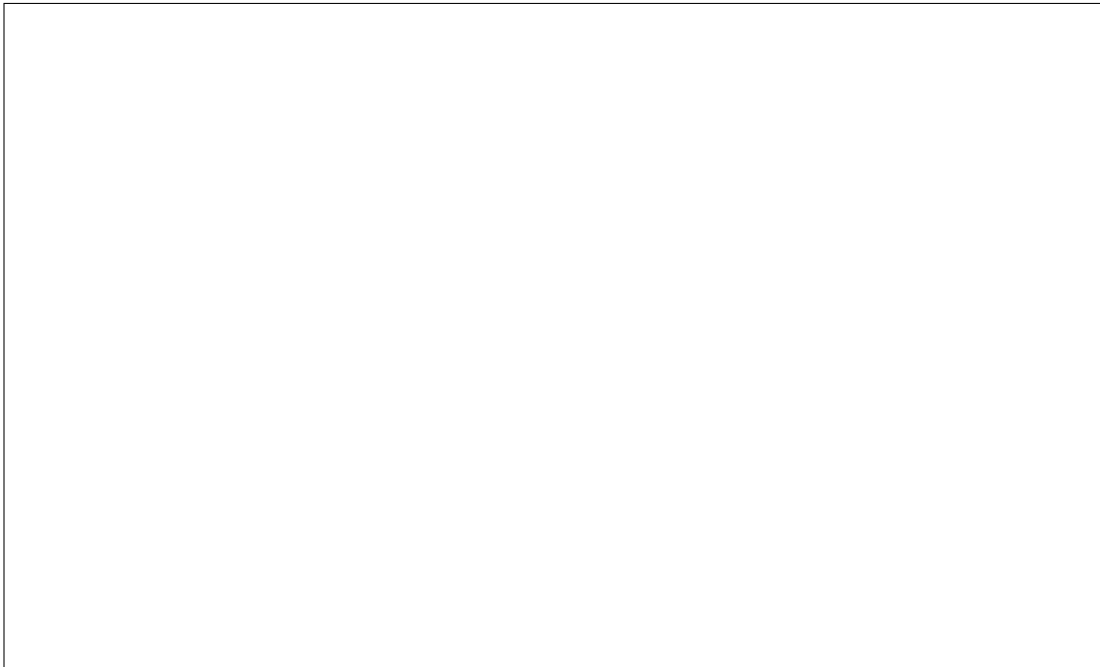


Figure 3: (Left) A camera on an arm is recording a video as it rotates around an axis. (Right) top view of the set up.

4. (12 points) LIGHT FIELD

Consider Figure 3. On the left you see a camera attached to an arm that is rotating around an axis. The camera is recording a video sequence as it is rotating. On the right, you see a schematic top view of the setup. Suggest a way to synthesize a novel view from the virtual camera inside the circle, using the images taken by the rotating camera.



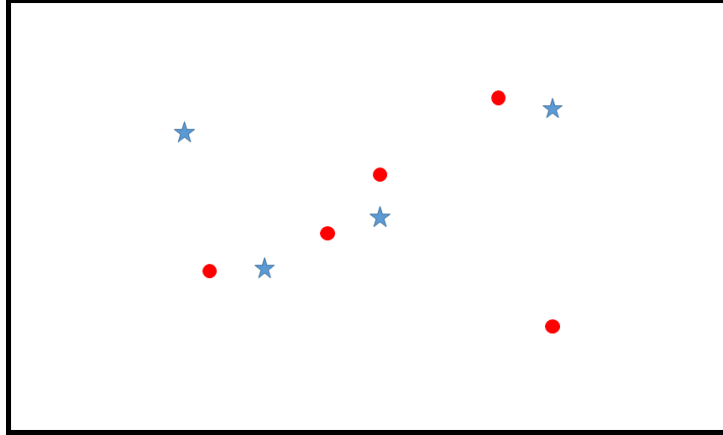
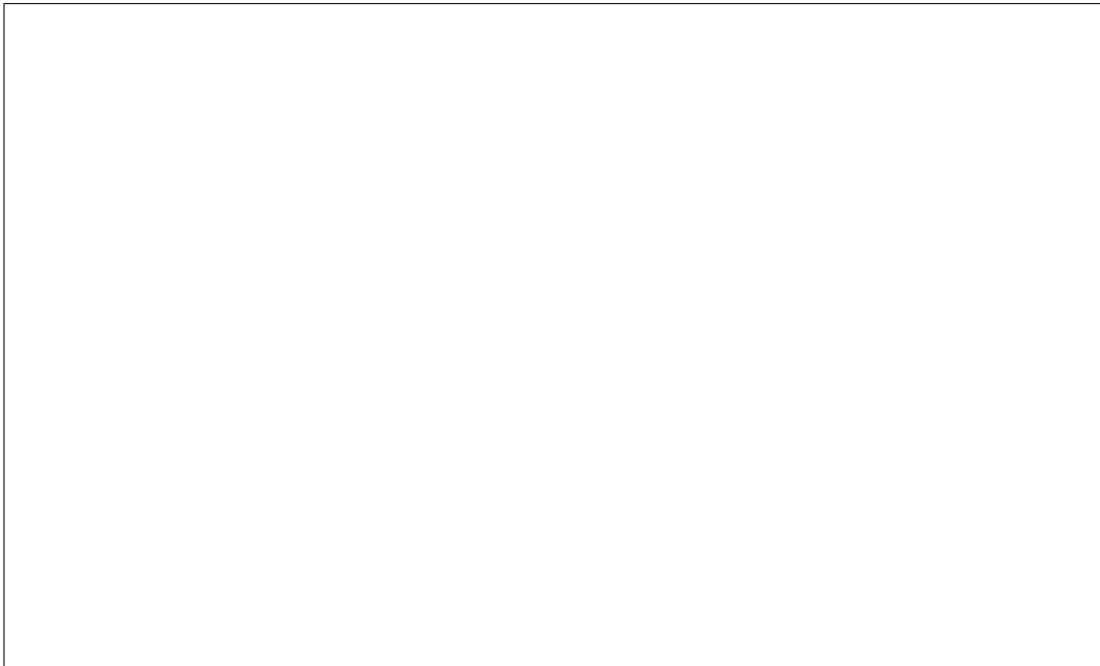


Figure 4: Blue stars represent predicted target positions. Red circles represent the measurements. You need to determine the likelihood of a measurement belonging to a particular target.

5. (13 points) TRACKING

Consider a multi-person tracking system. At each time step you have n targets. Target i is tracked using a Kalman Filter with state vector X_i and covariance matrix P_i . At each time step you obtain m measurements $Z_1, \dots, Z_m$ (say by running a person detector). Both the state X and the measurement Z are 2D vectors that represent the 2D position of a target. See Figure 4. Design an association matrix Ω of size $n \times m$, where $\Omega(i, j)$ determines how likely is measurement j to belong to target i . Express $\Omega(i, j)$ in terms of the Kalman filter parameters.



6. (12 points) COVID-19

The Covid-19 pandemic spreads at an exponential rate. The measured daily number of new cases, Z_t , can be written as $Z_t = e^{-(r_t+n)}$ for some daily growth rate r_t and white Gaussian noise n . Experts tell us that r_t can be modeled as $r_t = r_{t-1} + \Delta r_{t-1}$. Design an on-line model that at day t predicts Z_{t+7} , the number of new cases a week ahead.



Figure 5: (Left) An image with superpixels overlaid on it. (Middle, Right) zoom-ins.

7. (13 points) SEGMENTATION

Superpixels is a mid-level image representation that breaks the image into small and homogenous regions called superpixels. An image is usually represented by around 1000 or 2000 superpixels, as this reduces the computational burden of subsequent tasks. See Figure 5 for an example. Write a pseudo-code of an algorithm that uses geodesic distances to calculate superpixels of an image. Your algorithm should take as input the image and the number of requested superpixels.

8. (12 points) ANOMALY DETECTION

Machines in a factory make sound. This sound changes if the machine is about to break. It is therefore useful to record the sound of the machine and give a warning in case an abnormal sound is detected.

Your goal is to design such a system. Assume a microphone records the sound of the machine and produces a feature vector $X_t \in R^d$ every second. You know that the signal X over time fits a Gaussian Mixture Model with $k = 3$ Gaussians. Suggest a method to detect anomalies in the sound of the machine.